

— SOURCING & INTEGRATION · COMPREHENSIVE REPORT

Sourcing & Integration of Color E-Paper Displays

From panel to system — an integrated decision & engineering guide

Two tracks: **Indoor & Outdoor** · Procurement tiers: **Complete kit / Bare panel / Driver board**

Integration with **Raspberry Pi · ESP32/Arduino/STM32 · Android**

Verified specs, prices & sourcing channels — tailored for Doha's climate

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Benchmark: **Samsung Color E-Paper (EM32DX)**

Sources: **E Ink · Good Display · Geniatech**
Waveshare · IRIS Optronics ·
buyepaper · Samsung

© 2026 Abdulaziz Nasser AL-Haddad. All rights reserved. v10 Enhanced Edition — integrates the project TDD & deployment SOP (content-push protocol, hardware qualification, 16-week rollout, procurement data model, security considerations) with the full kiosk architecture, deployment timeline, and procurement ERD delivered as in-report figures.

COMPREHENSIVE REPORT · REVISION 10

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SECTION 01

Executive Summary & Decision At-a-Glance

01

This report consolidates a multi-stage investigation into sourcing and integrating **color E-Paper displays** as an alternative to the Samsung Color E-Paper — in **panel + board**, **bare panel**, or **standalone driver board** form — for two separate use cases: **indoor** and **outdoor** in Doha's climate. All figures and specifications below are drawn from official product pages and specialist sources.

Two decisions govern everything else: **(a) temperature** dictates the technology (indoor ≠ outdoor), and **(b) the panel's interface** dictates the driver board (large color panels require a matching TCON — there is no generic board).

TRACK	RECOMMENDED OPTION	TEMP	SUGGESTED CONTROLLER
Indoor	Good Display 31.5" Spectra 6 + ESP32-S3 board	0 → 50°C	ESP32 (embedded), or Android/RPi via WiFi/USB.
Outdoor	E Ink Kaleido 3 Outdoor 25.3" + Luvia board	-15 → 65°C	Official E Ink board + IP65 enclosure & shading.

THE FOUR KEY POINTS: ① Do not use a Spectra 6 panel (indoor, 0–50°C) outdoors in Doha. ② A large color panel (25.3"/31.5") requires the manufacturer's TCON — the generic IT8951 is grayscale only. ③ Android does not drive SPI directly; its role is a content source over WiFi/USB to the driver board. ④ **Sequence the purchase (§11):** nothing ≥25.3" is bought until a one-page content spec is written and at least two supplier quotes are in hand — the pipeline is proven first on a ~\$298 13.3" kit.

SECTION 02

Governing Principles

02

2.1 Principle one: temperature separates the two tracks

- **Spectra 6** (most vivid color, 6-particle): operates **0–50°C** → indoor. Doha summers reach 45–50°C ambient, and a panel surface in direct sun hits 60–70°C → risk of ghosting and permanent damage outdoors.
- **Kaleido 3 Outdoor**: **–15 to 65°C** → practical outdoor, sunlight-readable, solar-capable (muted color, color-filter technology).
- **Marquee**: **–20 to 65°C**, vivid outdoor color, but 55" and partner-only.
- **ChLCD** (IRIS Optronics): **–20 to 70°C**, the most robust color in direct sun, but a B2B solution.
- **"Indoor" is not always <50°C in Qatar**: glass atriums, full-height-glazed lobbies, and window-adjacent retail can reach 55–60°C in direct sun — beyond Spectra 6's range. Treat such sun-exposed indoor spots as outdoor for panel selection.

2.2 Principle two: the panel↔driver pairing rule

A panel is never wired directly; it terminates in a fragile FPC ribbon that plugs into an adapter/driver board, which then connects to the host. The driver board type is set by the panel's interface:

- **Small SPI (≤13.3")**: a cheap adapter (DESPI / Waveshare HAT) + your own controller (RPi/ESP32/STM32). Flexible and economical.
- **Large color (25.3" Mini-LVDS / 31.5" QSPI 60-pin)**: the manufacturer's matching TCON only — no generic board exists.
- **Large grayscale Carta**: the generic IT8951 (USB/SPI/I80) — but black & white, no color.

Practical rule for Doha

Even outdoor-rated panels (–15/65°C) need an IP65 enclosure + sun-shade + anti-glare, because direct sun can push the panel surface above 65°C in summer. Refresh also slows in high heat.

2.3 Design goals & non-goals (from the project technical design document)

The kiosk project's TDD fixes the requirements that every selection in this report must serve — and, just as importantly, what the system deliberately does *not* attempt:

GOALS

- **G1 Environmental resilience** — reliable operation across $-20 \rightarrow +60^{\circ}\text{C}$ ambient, IP65 for outdoor sites.
- **G2 Offline-first** — full function with no network: local rendering and caching.
- **G3 Power efficiency** — zero standby power on the display subsystem (bistability).
- **G4 Procurement traceability** — structured RFQ $\rightarrow$ deployment workflow with audit trail (§11.5).
- **G5 Content integrity** — CRC validation + staleness detection on every push (§7.5).
- **G6 Prototype-first** — pipeline proven on a low-cost kit before any production panel is ordered (§11.2).

NON-GOALS

- **NG1** No video or animation — e-paper refresh is inherently slow; designing for it invites failure.
- **NG2** No cloud dependency — the system must be complete offline; cloud is an attack surface and a liability, not a feature.
- **NG3** No multi-vendor panel mixing within one deployment — one panel type per site keeps waveforms, spares, and procedures uniform.
- **NG4** No blanket color — color panels only where a genuine stakeholder requirement exists (the §11.1 decision rule).

SECTION 03

Color E-Paper Technology Comparison

03

TECHNOLOGY	COLOR	TEMP	AVAILABLE AS PANEL+BOARD	BEST FOR
Spectra 6	6-particle — most vivid (60,000-color gamut)	0→50°C	Readily available (Good Display)	Indoor color
Kaleido 3 Outdoor	4096 (color-filter) — muted	-15→65°C	Available (+ Luvia/CONCERTO)	Practical outdoor color
ChLCD (IRIS)	Accurate, vivid color	-20→70°C	B2B / contact	Harsh outdoor (direct sun)
Marquee	4-particle — vivid	-20→65°C	Partner-only (55")	Large vivid outdoor
Carta	Black & white (16 grays)	Wide	Available (+ IT8951)	Economical outdoor, no color

Note: Spectra 6 won SID's Display of the Year award; E Ink is the world's largest ePaper supplier (sources: E Ink, SID). Temperature figures are from official product pages.

SECTION 04

Track 1 — Indoor Use (color)

04

All use **E Ink Spectra 6** (the most vivid), at **0W** static and **under 6W** during updates. The closest size to the Samsung 32" is **31.5"**.

PRODUCT	SIZE / RESOLUTION	BOARD / INTERFACE	APPROX. PRICE	NOTE
Good Display GDEP315C01(E6)	31.5" / 2560×1440	+ ESP32-S3 board (WiFi/BT/TF)	~\$700–1,200	Closest to Samsung. QSPI 60-pin. PMIC.
Good Display GDEP253C02(E6)	25.3" / Mini-LVDS	+ TCON + Android board (USB)	~\$1,000–1,400	Easy USB updates (copy images).
Good Display GDEP133C02(E6)	13.3" / 1600×1200	+ ESP32-133C02 (WiFi/BLE/USB/SD)	~\$300–449	Built-in web server. Ideal for prototyping.
E Ink Online Shop (direct)	25.3" / 13.3" / 7.3"	+ separate driving board	\$1,400 / \$449 / \$149	31.5" currently out of stock there.
Waveshare	up to 13.3"	+ ESP32-S3 / HAT	4": \$50 · 13.3": ~\$298–327	Cheapest; does not exceed 13.3".

RECOMMENDATION: **Good Display 31.5" Spectra 6 + ESP32-S3** — the direct equivalent of the Samsung display. For prototyping/budget: Waveshare 13.3" (~\$298). For a direct purchase with factory warranty: E Ink's shop (25.3" at \$1,400).

Sources: good-display.com, buyepaper.com, shopkits.eink.com, Waveshare.

Hot-indoor caveat (Qatar)

Spectra 6's 0–50°C range leaves no margin in sun-exposed interiors — glass atriums, glazed lobbies, west-facing windows — where Doha indoor temperatures can reach 55–60°C. For those locations choose an outdoor-rated technology (Kaleido 3 Outdoor / ChLCD) or a wide-temp B&W panel, not Spectra 6.

SECTION 05

Track 2 — Outdoor Use (Qatar Heat)

05

Color options are narrower here. The practical panel+board choice is **Kaleido 3 Outdoor**; for the most robust color in sun, **ChLCD**; for the largest/most vivid via partners, **Marquee**.

PRODUCT / TECHNOLOGY	SIZE	FORM / BOARD	TEMP	NOTE
E Ink Kaleido 3 Outdoor	25.3"	+ Luvia board (separate)	-15→65°C	Best in practice. Muted color (filter). Solar.
E Ink Kaleido 3 Outdoor	13.3"	+ CONCERTO board (panel ~\$449)	-15→65°C	For smaller outdoor prototypes.
Geniatech EPC2530 (Kaleido 3)	25.3"	Semi-finished unit (CMS, 4G, battery)	Outdoor	3200×1800 native; effective color res lower (color-filter). Sunlight-readable, remote mgmt.
IRIS Optronics — ChLCD	Signage	B2B solution (no standalone board)	-20→70°C	Most robust, accurate color in sun + IP56.
E Ink Marquee (4-particle)	55"	Partner-only	-20→65°C	Large vivid color; not panel+board; new.
Wide-temp B&W 31.2" (GDEP312TT3 / DMPH312EC1 / E Ink V32-OSM)	31.2"	Bare panel, or IP65 outdoor unit	-15→65°C	2560×1440, 16-gray. Robust & low-cost, but B&W (no color).
SEEKINK (S315E6 / S280EC-O)	28"–31.5"	Billboard / outdoor-rated	incl. -15→65°C	Alternative supplier: Spectra 6 billboards + outdoor models.

RECOMMENDATION: **Kaleido 3 Outdoor 25.3" + Luvia board** (balance of color/temp/availability). For more accurate, robust color → **ChLCD**. Large & vivid via partners → **Marquee**. Color not required → **wide-temp B&W 31.2"** (GDEP312TT3, or the IP65 V32-OSM / DMPH312EC1).

Sources: shopkits.eink.com (Kaleido Outdoor + Luvia/CONCERTO; V32-OSM), eink.com, good-display.com (GDEP312TT3/DMPH312EC1), geniatech.com, iris-opt.com, seekink.com.

SECTION 06

The Three Procurement Tiers

06

6.1 Complete kit (panel + board)

Fastest to deploy: panel + matching driver board + code & docs. This is the practical route for large color panels (see the two track tables). Example: Good Display 31.5" Spectra 6 + the ready ESP32-S3 board with WiFi/BT/TF.

Why choose a complete kit? The manufacturer has already solved the hardest integration problem: matching the panel's specific waveform requirements to the TCON's output. You get working firmware, example code, and a warranty that covers the pair.

6.2 Bare panel (panel only) — when does it make sense?

- **Small ($\leq 13.3"$ SPI):** sensible and economical — bare panel + a DESPI adapter + your own controller. Example bare-panel prices: Waveshare 4" ~\$49.52, 13.3" ~\$326; E Ink: 7.3" ~\$149, 13.3" ~\$449.
- **Large color (25.3"/31.5"):** "panel only" is not a real saving — it still needs the manufacturer's TCON (no generic board); suitable only if you are designing your own driver board (PMIC + waveforms).
- **Caution:** the FPC ribbon is fragile (bend only along the horizontal), and the panel runs at 3.3V, so a level shifter is needed with a 5V Arduino.

6.3 Standalone driver boards — catalog by category

CATEGORY / EXAMPLE	DRIVES	HOST CONNECTION	NOTE
DESPI adapters (C02/C73/C1085...)	Small SPI panels $\leq 13.3"$	SPI (GPIO) + your MCU	Cheap ~\$5–20; DESPI-L includes USB serial.
ESP32 dev boards (ESP32-133C02, 31.5" board, Waveshare E6)	Specific Spectra 6 color panels	WiFi/BT/USB	The practical color route ~\$30–80; web server.
E Ink official (CONCERTO, Luvia)	Kaleido 3 (Outdoor) modules	Per design	Design support from E Ink; price on request.
IT8951 (generic)	Carta grayscale 6"–13.3"+	USB / SPI / I80	Black & white only (16 grays) ~\$40–70.

CATEGORY / EXAMPLE	DRIVES	HOST CONNECTION	NOTE
Waveshare HAT (40-pin)	Waveshare SPI panels	Raspberry Pi GPIO	Direct mounting on RPi.

Sources: *good-display.com (DESPI/ESP32-133C02), shopkits.eink.com (CONCERTO/Luvia), waveshare.com (IT8951/HAT).*

SECTION 07

Integration & Wiring (Hosts)

07

The panels work with **Raspberry Pi**, **ESP32/Arduino/STM32**, and **Android** — but the role and connection method differ:

7.1 Raspberry Pi (best compatibility)

- Small/medium SPI (≤ 13.3 " color): via a **HAT/DESPI** on the 40-pin header. Spectra 6 panels connect to RPi/Jetson/Arduino/STM32 through a driver HAT with a standard 40-pin header.
- Large grayscale (Carta): via **IT8951** over USB (or SPI/I80).

7.2 Arduino / ESP32 / STM32 (embedded control)

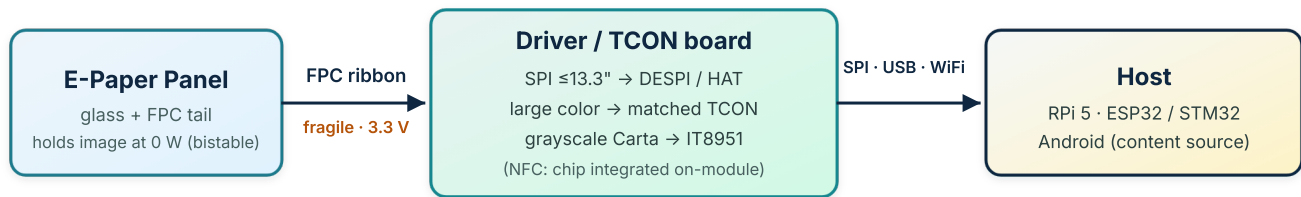
- SPI connection via **DESPI + FPC cable**; with STM32/ESP32/Arduino code from Good Display.
- **ESP32 is effectively the standard** — the color driver boards themselves are built on it (ESP32-133C02 for the 13.3" color: QSPI + WiFi/BLE/USB/SD).
- Caution: an Arduino Uno is memory-constrained for large color panels (large framebuffer) — use ESP32/STM32.

7.3 Android (the source — fits the kiosk)

- Android does not typically drive SPI directly (restricted GPIO access). Its role is a **content source** sending images to the driver board (the actual controller) via:
- **USB**: fast updates for Android devices, or copy images to a USB drive and process automatically.
- **WiFi**: a built-in web server you update from any phone/device on the same network.
- Flow: Android app → WiFi/USB → TCON board → panel.

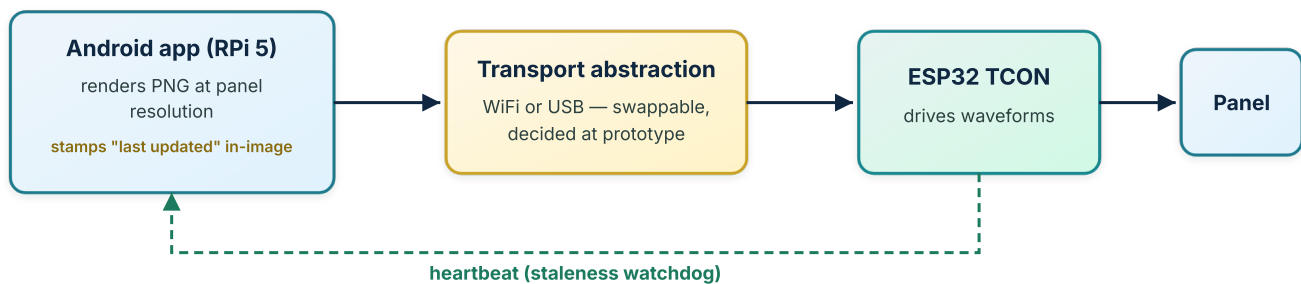
7.4 Connection points (the cable / custom part)

- Panel → flat **FPC** ribbon → adapter connector: lift the latch, insert the panel (white side up), close the latch.
- Small SPI: the **DESPI** series. Large color: the connector built into the matching TCON (e.g., 60-pin for the 13.3" QSPI).
- Then driver board → host via **SPI** (GPIO), **USB** (IT8951/Android), or **WiFi** (ESP32 boards).
- Caution: the FPC ribbon is fragile (bend only horizontally); 3.3V logic (level shifter for 5V).



A panel is never wired directly to the host — the driver board type is dictated by the panel's interface (§2.2, §6.3).

Figure 1 — Physical signal chain: panel → FPC → driver board → host.



Power outage: the panel keeps its last image at 0 W — only the controller and access logic need a UPS (§9).

Figure 2 — Kiosk content-push architecture: content logic separated from panel driving, hardened with the staleness watchdog.

Proposed architecture for the kiosk project (RPI5 + Android + ESP32)

Make the **driver board (ESP32 TCON or E Ink board)** the **direct controller of the panel**, fed by the **RPI5/Android as a content source over WiFi/USB**. This separates **content logic** (RPI/Android) from **panel driving** (TCON) — the same critical/non-critical separation principle you applied in the kiosk design. The result: reliable updates, easier maintenance, and fault isolation. Two hardening rules complete the architecture: **(a) formalize the interface** — render to a fixed intermediate (PNG at panel resolution) and push through a thin transport abstraction, so WiFi vs USB stays a deferred, cheap choice and the panel remains swappable without touching app logic; **(b) add a staleness watchdog** — bistability means a wedged TCON keeps showing yesterday's content convincingly, so the TCON must emit a heartbeat and the app must render a "last updated" timestamp into every pushed image.

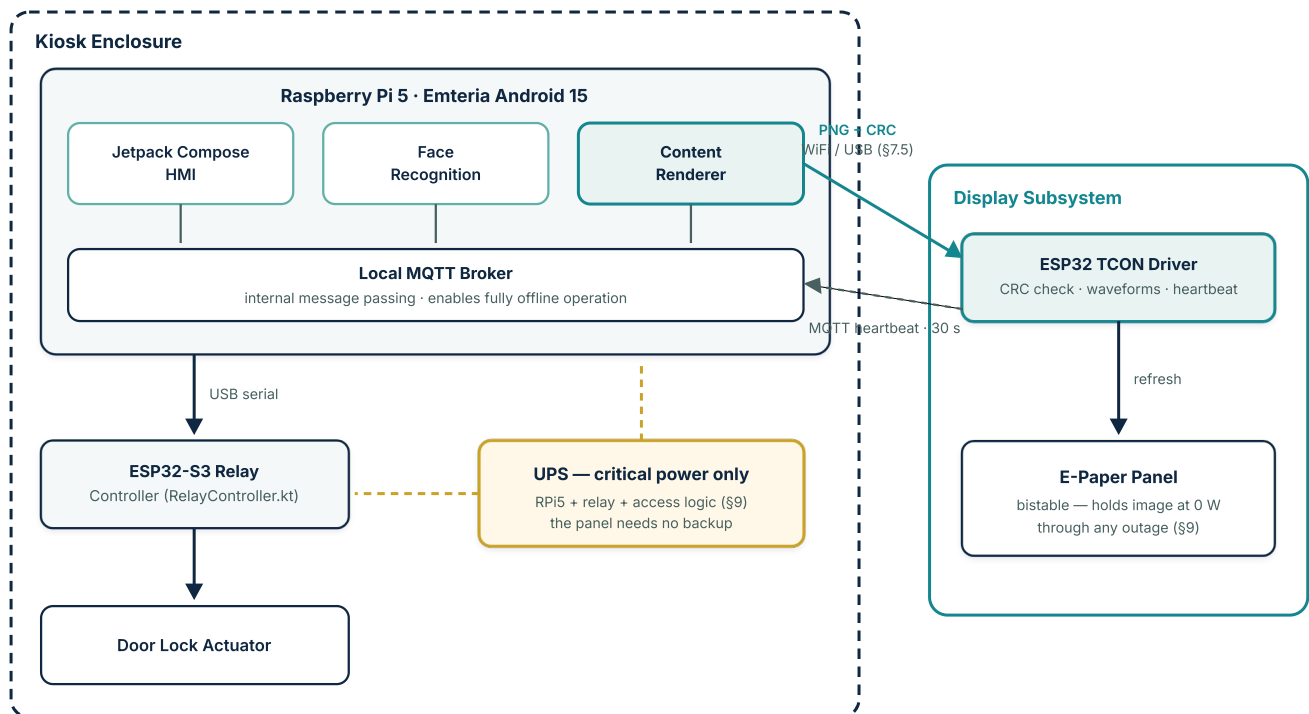


Figure 10 — Full kiosk system architecture (from the project TDD): the §7 callout as a system view — access-control HMI, content rendering, and panel driving in separate layers; the UPS backs only the critical path, because the bistable panel keeps its image at 0 W.

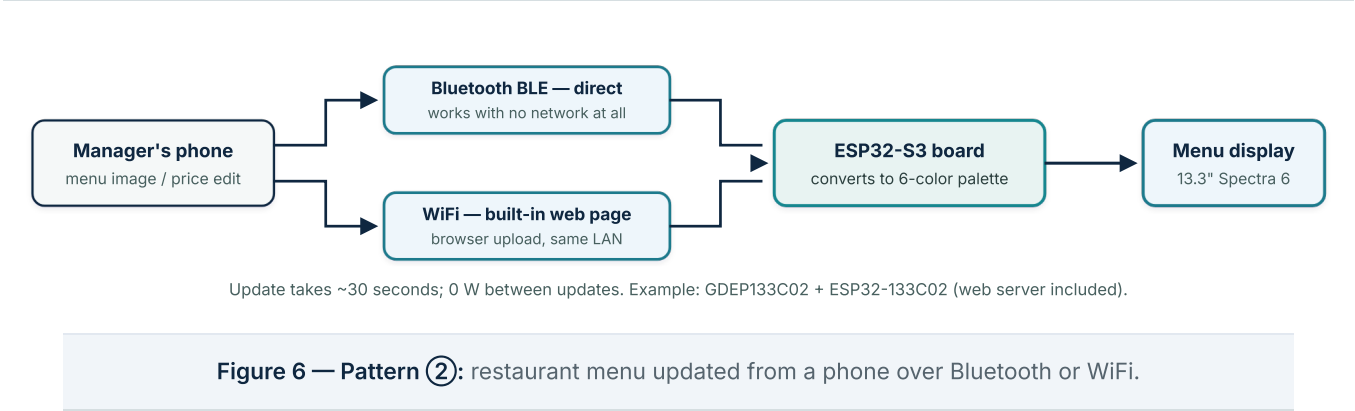
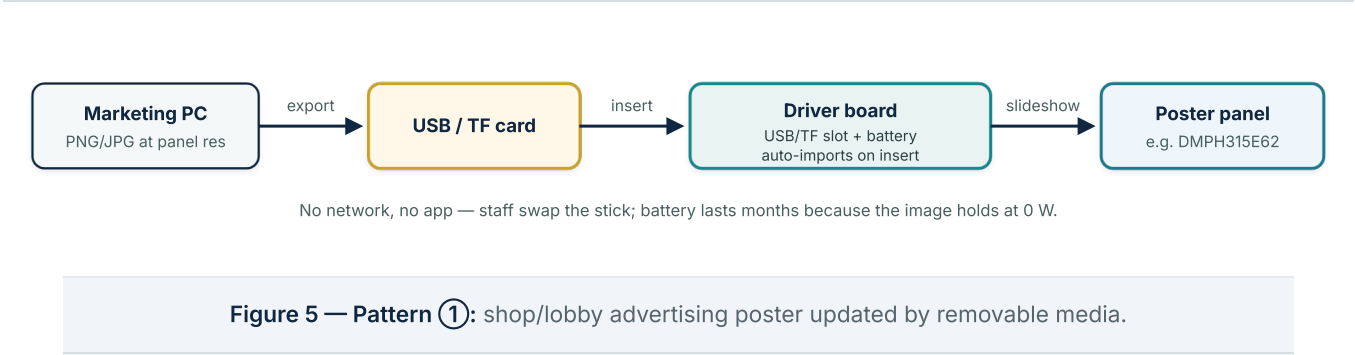
Application patterns — five ways the image gets onto the panel

Choosing the *update channel* is as architectural a decision as choosing the panel itself. Five proven patterns cover the practical spectrum, from fully manual to fully automatic.

PATTERN	TYPICAL PROJECT	HOW THE IMAGE ARRIVES	HARDWARE EXAMPLE	CHOOSE WHEN
① Removable media (USB / TF card)	Shop or lobby advertising poster	Staff export PNG/JPG at panel resolution to a flash stick; the board auto-imports on insert and plays a slideshow	DMPH315E62 finished billboard; GDEP253C02 + Android TCON board	Updates are occasional, staff are on-site, and no network is wanted
② Phone push (Bluetooth / WiFi)	Restaurant menu, info board	Phone app over BLE (works with no network at all), or a browser upload to the board's built-in web server on the same LAN	GDEP133C02 + ESP32-133C02 (web server included)	Daily price/content edits from a phone, no PC in the loop
③ NFC tap (battery-free)	Door labels, shelf tags, name badges	A 2–5 second tap delivers <i>both</i> harvested power and the image over the same 13.56 MHz field (\$10)	Waveshare NFC modules 1.54"–7.5"; ST25R3911B reader for bulk programming	Small B&W labels with zero wiring and zero batteries

PATTERN	TYPICAL PROJECT	HOW THE IMAGE ARRIVES	HARDWARE EXAMPLE	CHOOSE WHEN
④ Cloud CMS (4G / internet)	City-wide outdoor signage fleet	A central CMS schedules content per site and pushes over 4G; units return battery/temperature telemetry; solar + battery power	Geniatech EPC2530-class units with remote CMS	Many remote displays needing central control and health monitoring
⑤ Self-generating host	Live dashboard — and this report's kiosk	The host renders PNG from data sources (APIs, schedules) on a timer and pushes it through the transport abstraction; the in-image timestamp guards freshness	RPi5 + ESP32 TCON — exactly Figures 1–2	Content derives from data; zero-touch operation

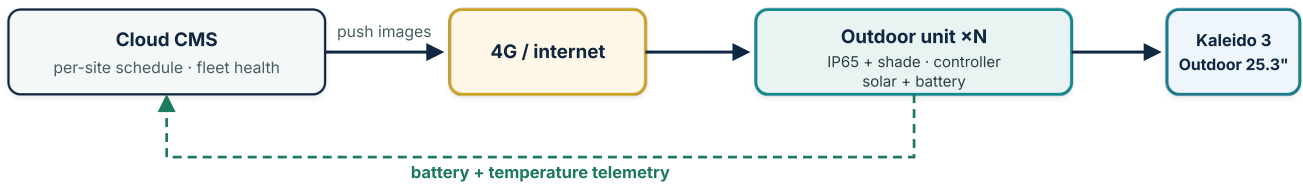
Patterns ①→⑤ trade manual effort for autonomy. The kiosk project is pattern ⑤; patterns ①–④ are reusable recipes for sibling deployments (ads, menus, labels, fleets) on the same panel knowledge in this report.





Sizes 1.54"–7.5", B&W (\$10). Bulk programming: ST25R3911B reader board — works with all phones' tags.

Figure 7 — Pattern ③: battery-free door labels and badges updated by an NFC tap.



Solar easily sustains it — content shows at 0 W between updates; telemetry catches overheating and stale units.

Figure 8 — Pattern ④: city-wide outdoor fleet managed remotely from a central CMS over 4G.



Zero-touch: nobody uploads anything — content updates itself. This is the kiosk architecture of Figures 1–2 and 4.

Figure 9 — Pattern ⑤: live information dashboard (reception/mosque) with fully automatic image generation.

7.5 Content-push protocol — formal specification (from the project TDD)

The handshake of Figures 2 and 4 is pinned down in the project's technical design document as an implementable wire protocol. The Android renderer pushes the image over HTTP to the ESP32 TCON; status and heartbeats return over the local MQTT broker already running on the RPi5 — no cloud involved.

Image push (Android → TCON, HTTP):

```

POST /display/update
Content-Type: application/octet-stream
X-Image-CRC: <CRC-32 hex>
X-Image-Width: <pixels>
X-Image-Height: <pixels>

<PNG binary data>
  
```

RESPONSE	MEANING
200 OK	Image received, CRC validated, refresh initiated.
400 Bad Request	CRC mismatch or malformed image — the app must re-render and re-push, never retry the same bytes blindly.
503 Service Unavailable	TCON busy with a previous refresh (tens of seconds on large panels, slower in heat §12) — back off and retry.

Refresh-complete event (TCON → broker, MQTT display/status):

```
{ "event": "refresh_complete",  
  "duration_ms": 2400,  
  "panel_temp_c": 42 }
```

Heartbeat (TCON → broker, MQTT display/heartbeat, every 30 s):

```
{ "timestamp": "<ISO-8601>",  
  "uptime_s": 86400,  
  "last_refresh_crc": "A1B2C3D4" }
```

Staleness rules: heartbeat interval 30 s; alarm on **3 consecutive missed heartbeats (90 s)** — raise the TCON-stale alarm, log the event, attempt reconnection (the Figure 4 alarm branch). Two useful by-products of this payload design: panel_temp_c doubles as the §12 thermal telemetry, and last_refresh_crc lets the app verify *what is actually on glass*, not just what was last sent.

SECTION 08

Indicative Price Table (BOM) by Size

08

Estimates to guide budgeting; confirm with the supplier (panel and board are usually separate).

SIZE	TECHNOLOGY	BARE PANEL	MATCHING DRIVER BOARD	COMPLETE KIT
4"	Spectra 6	~\$49.52 (Waveshare)	Adapter/HAT ~\$10	~\$60
7.3"	Spectra 6	~\$149 (E Ink)	DESPI-C73 ~\$10–15	~\$120–160
13.3"	Spectra 6	~\$326 (Waveshare) / \$449 (E Ink)	ESP32-133C02 ~\$40–70	~\$298 (Waveshare bundle)
25.3"	Spectra 6	~\$1,400 (E Ink retail)	Factory TCON (required)	~\$1,000 (ODM)–1,400
31.5"	Spectra 6	~\$580–600 (Good Display)	ESP32-S3 board (required)	~\$700–1,200
13.3" (outdoor)	Kaleido 3 Outdoor	~\$449 (E Ink)	CONCERTO (separate)	On request
25.3" (outdoor)	Kaleido 3 Outdoor	On request	Luvia (separate)	On request
31.2" (grayscale)	Carta B&W, –15→65°C	~\$580 (GDEP312TT3)	IT8951 ~\$40–70	IP65 unit (DMPH312EC1 / V32-OSM) on request

Reading the columns: bare-panel figures are typically E Ink *retail*, while complete-kit figures are *ODM* (Good Display / Waveshare) — so an ODM kit can legitimately cost less than E Ink's retail bare panel. This is a cross-supplier difference, not a pricing error; confirm each figure and package contents with the supplier. Add an IP65 enclosure & shading to outdoor budgets, and ~5% customs duty on imports into Qatar.

SECTION 09

Power-Outage Behavior & Image Retention

09

All of the displays in this report are bistable electronic paper: the pigment particles physically hold their position once set, so the panel draws power **only while the image is changing** and **zero power to keep an image on screen**. There is no backlight to lose. This makes e-paper uniquely suited to power outages and off-grid sites — the opposite of LCD/LED/OLED, which go black the instant power is removed.

Key Questions Answered

- **Does the image stay visible when power is cut?** Yes, on every display here — the last image freezes and remains fully readable with zero power, **indefinitely** (months to years).
- **Can it operate during an outage?** To *show* the last image: always. To *keep updating* content: a power source (battery or solar) is required, because a refresh needs power.
- **How long without electricity?** Displaying the frozen image: effectively unlimited. Continued updates: battery/solar-dependent (see table).

DISPLAY	LAST IMAGE STAYS W/O POWER?	POWER SOURCE	KEEPS UPDATING OFF-GRID FOR...
Samsung EM32DX (32" reference)	Yes — indefinitely	Built-in 4,600mAh + USB-C	~200 days @ 1 update/day; up to ~500 days if rarer
Good Display 31.5" billboard (DMPH315E62)	Yes	Built-in lithium battery	Untethered; scales with update frequency
Panel + ESP32-S3 board	Yes	Battery header (add LiPo)	Months at low update rates
Waveshare 13.3" + ESP32-S3	Yes	Onboard lithium-battery header	Depends on your LiPo
Bare panel only (any size)	Yes	None (you supply)	Image persists forever; updating needs your power
Kaleido 3 Outdoor (25.3"/13.3")	Yes	Solar-capable (external)	Long-term on solar — depends on panel/battery sizing & cleaning (Qatar dust)
Geniatech EPC2530 (outdoor 25.3")	Yes	External battery option	Extended off-grid with battery

DISPLAY	LAST IMAGE STAYS W/O POWER?	POWER SOURCE	KEEPS UPDATING OFF-GRID FOR...
NFC battery-free e-paper tags	Yes	None — harvests RF from the reader	No stored energy at all; image persists indefinitely (smaller sizes)
Marquee / ChLCD / Carta	Yes (all bistable)	Per your integration	Per power source

- **"0.00W" is the static figure** — technically under 0.005W, reported as 0.00W per IEC standards. A refresh briefly draws a few watts, and battery runtimes assume a fixed update cadence (Samsung's 200 days = one update/day; optional battery kits on AC models last roughly 3–6 weeks at 4 updates/day). Frequent updates shorten runtime sharply.
- **Critical for the kiosk:** the panel retaining its image during an outage does *not* mean the system keeps working. The e-paper keeps showing the last screen, but the controller (RPi5 / ESP32) and any door/access logic still need power to read a card, respond, or push a new image. Use e-paper to keep *information* visible through an outage; give interactive/control functions their own UPS or battery.
- **Heat shortens battery life (not image retention):** in Doha's heat, lithium-battery runtime degrades faster, so battery-powered units may fall short of their rated days — but the bistable image itself is unaffected and still holds at zero power. Solar, NFC, or wired options sidestep this.
- **UPS sizing is currently unsizeable:** the peak power draw of a large-panel refresh (25.3"/31.5") is not published — request it as an RFQ line item or measure it on the 13.3" prototype before specifying the kiosk UPS (§11).

BOTTOM LINE: for power-unreliable contexts, e-paper is ideal — whatever was last displayed stays readable for the entire outage at no power, and battery/solar units can even keep refreshing.

Sources: Samsung (EM32DX user manual, Newsroom), E Ink, Good Display, Geniatech.

SECTION 10

NFC E-Paper: Battery-Free, Phone/Reader-Updated Displays

10

NFC e-paper is a distinct branch that needs **no battery and no separate driver board**. The NFC chip on the module *is* the driver, and it harvests its power wirelessly from whatever updates it — there is no SPI host, no TCON, and no wiring. You tap an NFC phone (or hold an NFC reader) near the module; the same RF field both powers the chip and transfers the new image.

Where it sits in the driver taxonomy (§6): unlike the SPI adapters (DESPI), the color TCONs (ESP32-133C02 / Luvia), or the grayscale IT8951, an NFC module is fully integrated — the "driver board" and the "host" collapse into (a) an on-module NFC+EPD chip and (b) an external NFC field source (phone or reader).

10.1 How it works

Bring an NFC phone/reader within a few centimetres of the module's antenna → the RF field powers the chip and carries the image → the refresh finishes in a few seconds → the image then stays with **zero power, indefinitely**.

10.2 Two typical workflows

- **Update the image from your phone's gallery:** in the module's Android app, pick a photo (or shoot one), crop/resize/rotate, adjust brightness and contrast, then tap the phone to the module to push it.
- **An e-paper name/business card updated by tap:** the same flow, but the content is a card design (name, title, logo, QR) or a text image made in the app. Card-like sizes: 2.13" (250×122) or 2.9".
Important distinction: here NFC is the *update* channel (you tap to change what is shown). If instead you want a *visitor* to tap the card and save your contact (vCard), that is a different function requiring a standard NDEF tag — do not conflate the two.

10.3 Available NFC e-paper (Waveshare)

Sizes: 1.54", 2.13", 2.7", 2.9", 4.2", up to **7.5" (800×480)**. Black & white (some small 3-color), low resolution, few-second refresh. Intended for price/asset/shelf labels (ESL), name badges, and small info tags; all keep the last image with no power.

10.4 NFC devices compatible with e-paper

DEVICE TYPE	EXAMPLE	NOTES
NFC smartphone	Android + the module's app	Easiest; powers + transfers on tap. The vendor app excludes some phones (Samsung/Google/Sony) and iPhone access is limited. Fine for small sizes.
Dedicated NFC reader board	Waveshare ST25R3911B NFC Board / Kit	The matched reader — marketed as ideal for refreshing passive NFC e-papers; ~1.4W output, onboard MCU, standalone or host-integrated. Recommended for larger sizes (4.2"/7.5") and for automation.
High-power HF reader IC/module	ST25R3916, TI TRF7970A, NXP PN5180	Workable if RF output is high enough and the protocol matches (NFC Forum Type 2/4/5, ISO 14443/15693). Verify per module.
USB NFC reader (PC)	Desktop reader/writer	Batch programming from a computer; must be high-power and protocol-matched.
Industrial NFC writer / handheld	ESL encoders	For programming many labels in retail/asset deployments.

Key requirement

The reader must deliver **enough RF output power** to both harvest-power and drive the EPD (the ST25R3911B's ~1.4W is sized for this) — low-power tag readers may fail, especially on larger panels. The module's protocol/commands must also match, so the vendor app or the matched reader is the guaranteed path.

FIT FOR THIS PROJECT: NFC e-paper maxes out around **7.5" and is B&W** — harvested power cannot refresh a large or color panel, so it is **not** an option for the 25"/31.5" color signage. Its place is small **battery-free auxiliary labels** (asset tags, room/desk labels, name badges) around the kiosk, and as the zero-stored-energy extreme of the retention behavior in §9.

Sources: Waveshare (NFC e-paper module wikis; ST25R3911B NFC Board), STMicroelectronics (ST25R3911B).

SECTION 11

Procurement Roadmap: Requirements First, Prototype, then Purchase

11

The flagship options in Tracks 1–2 should not be ordered straight from the tables: the outdoor flagship is quote-only, the deciding requirements are usually unwritten, and the large color panels are expensive enough that sequencing matters. This section turns the tracks into a purchase plan.

— 11.1 Step 1 — write the one-page content spec

One page decides the central trade-off, and it must exist before any panel ≥ 25.3 " is ordered. It states: **content type** (menu / wayfinding / status / promotional), **update cadence** (per hour / day / week), **viewing distance**, and the **install site's measured thermal reality** (not assumed — survey the actual spot, especially glass-fronted interiors per §2). The decision rule it feeds: if cadence is low and the content is informational, the wide-temp B&W **GDEP312TT3** (~\$580, $-15 \rightarrow 65^\circ\text{C}$) covers *both* tracks with one SKU and halves the budget; the color dual-track stands only if color is a genuine stakeholder requirement (e.g., brand/showcase value), not an inherited assumption.

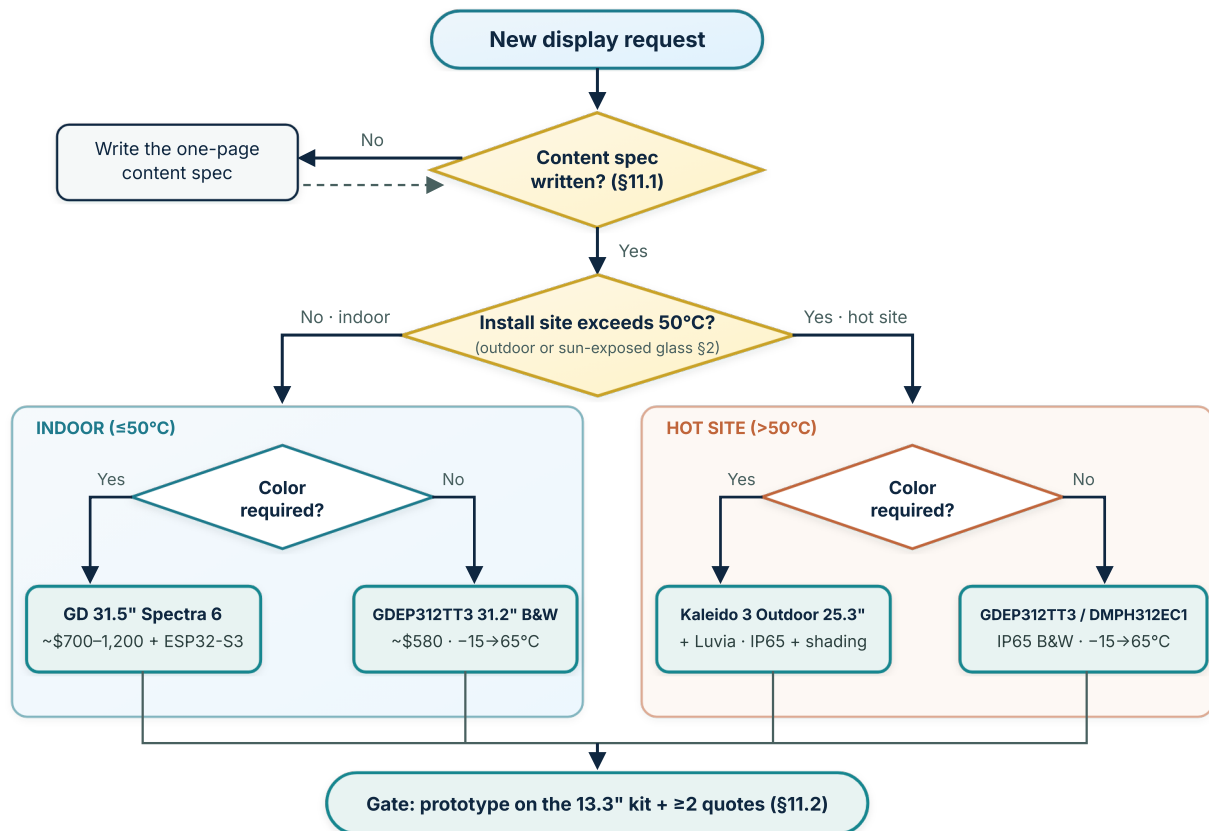


Figure 3 — Panel-selection decision tree: the content spec is the first gate; site temperature and the color question pick among four outcomes, and every path ends at the prototype-and-quotes gate.

11.2 Step 2 — prototype and quotes, in parallel

PHASE	ACTION	COST / TIME	OUTPUT
P0 — now	Sequence diagram of the Android→TCON content push, with link-down, power-outage, and staleness states made explicit	< 1 hour	Delivered ✓ — Figure 4 below; editable master in the companion pack (Appendix)
P0 — now	Three identical-spec RFQs: Good Display (GDEP315C01 + driver board / finished DMPH315E62 / GDEP312TT3), E Ink (25.3" Kaleido 3 Outdoor + Luvia), SEEKINK (S315E6 / S280EC-O)	1–2 weeks wait	Supplier × price × lead-time matrix
P1 — this month	Waveshare 13.3" Spectra 6 + ESP32-S3 kit: build the <i>full</i> pipeline (Android app → transport → ESP32 → panel), measure refresh time and 6-color conversion, practice FPC handling on a cheap panel	~\$298	Working pipeline; every line of code transfers to the final panel
P2 — last	Final panel purchase per the content spec + at least two quotes; GDEP312TT3 remains the budget fallback	per quote	Committed order with eyes open

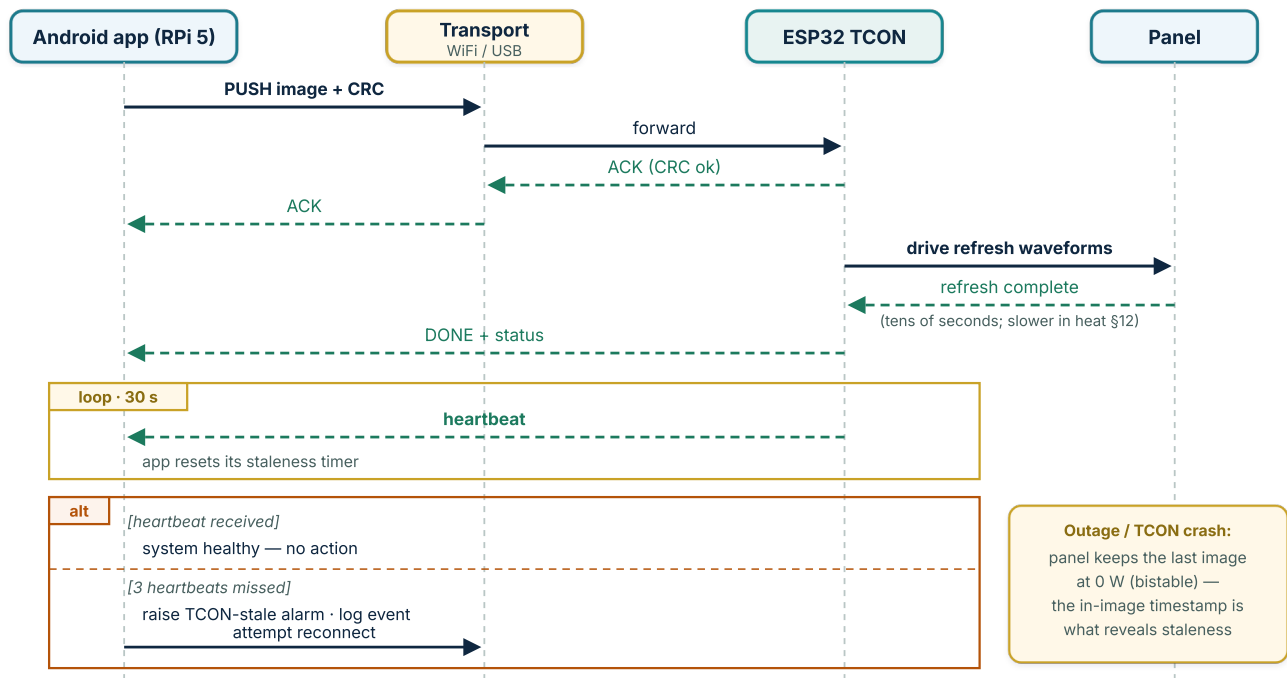


Figure 4 — The delivered P0 artifact: the Android→TCON content-push handshake, the 30-second heartbeat loop, and the 3-missed-heartbeats alarm branch. CRC = integrity checksum appended to each push; ACK = the receiver's confirmation.

Mandatory RFQ line items: unit price · lead time to Doha · MOQ · warranty terms *and DOA policy including return logistics to Qatar* (returning a dead 31.5" panel can cost more than the panel) · Incoterms (request DDP Doha) · **peak refresh power draw** (unpublished; needed for UPS sizing) · and for finished units: **is the content-push API open**, or locked to the vendor app? (An app-locked unit breaks the \$7 content interface.)

11.3 Integration hardening checklist

- **Staleness watchdog:** TCON heartbeat + "last updated" timestamp rendered into every pushed image (see §7) — the cheap fix for e-paper's unique failure mode of displaying stale content convincingly.
- **ESD & FPC handling protocol:** Doha's <30% RH indoor air makes ESD a genuine panel-killer. Wrist strap, grounded mat, written insertion procedure; practice on the 13.3" before touching a ~\$600 panel.
- **Acceptance test at delivery:** a written checklist run within the DOA window — power-on, full-white, full-black, color bars, ghosting check after 3 refresh cycles. For quote-only B2B imports this is the cheapest insurance available.
- **Content-rotation policy:** alternate at least two images on long-static deployments; static content held for weeks in heat invites ghosting at zero benefit.
- **Site thermal survey:** log the actual install location's temperature across a hot day before final panel selection (feeds Step 1).
- **Hardware qualification (production panels, from the project TDD):** temperature cycling $-20 \rightarrow +60^{\circ}\text{C}$ ×10 cycles in an environmental chamber; 10,000-refresh accelerated life test with no degradation; bench measurement of peak refresh power against the datasheet (this is the number the RFQ asks for and the UPS sizing in §9 is waiting on); water-jet IP65 validation for outdoor variants.

- **Acceptance test — concrete pass criteria (from the deployment SOP):** beyond the visual checks, push a test image and require 200 OK with a matching X-Image-CRC; read panel_temp_c from the heartbeat and require agreement with ambient within $\pm 3^{\circ}\text{C}$; record refresh power draw and the panel serial + firmware version in the deployment log. **FAIL → quarantine the panel and open the supplier RMA** — this is exactly why §11.2 demands the DOA policy as an RFQ line item.
- **7-day burn-in before handover (from the deployment SOP):** automated alerts on 3 missed heartbeats, panel temperature ($>55^{\circ}\text{C}$ indoor / $>65^{\circ}\text{C}$ outdoor), and UPS state; observe behavior through the 12:00–16:00 peak-heat window; push 3–5 different images across the week and check for ghosting; verify a deliberate TCON power-cut raises the staleness alarm at 90 s and recovers; UPS failover test with ≥ 15 min runtime; 72-hour fully-offline operation check.

11.4 Phased rollout plan (16 weeks, from the project TDD)

The §11.1–11.3 steps expand into a four-phase schedule in which engineering and procurement run *in parallel* and converge at a week-8 decision gate:

PHASE	WEEKS	ENGINEERING TRACK	PROCUREMENT TRACK	EXIT GATE
1 · Prototype	1–4	Content spec → sequence diagrams → order the 13.3" kit → build the pipeline, measure refresh timing	Identical-spec RFQ package → sent to 3 suppliers → await quotes (1–2 wk)	Working prototype pipeline
2 · Validation	5–8	FPC/ESD handling practice on the cheap panel; complete prototype validation; document lessons	Quotes in → supplier × price × lead-time comparison matrix	Readiness review: Spec + Quotes + Prototype — the decision gate
3 · Production order	9–12	—	Apply the Figure-3 decision tree per site → issue the PO → 4–6 wk typical lead → delivery acceptance test (§11.3)	Accepted panel(s) in hand
4 · Deployment	13–16	Site prep + enclosure (1 wk) → panel integration & wiring (2–3 d) → system integration tests (2–3 d) → 7-day burn-in → handover		Operations sign-off

Deployment governance: the field procedure is formalized as **SOP-EPAPER-DEPLOY-001** with a T-day timeline — site assessment & panel-selection verification at T–14...T–7, delivery acceptance test no later than T–7 (*before the panel ever travels to site*), installation at T-day under ESD discipline, integration testing at T+1, burn-in T+2...T+7, final acceptance and Operations handover at T+8 with a complete documentation package (site checklist, acceptance forms, wiring photographs, serials/firmware, burn-in summary).



Figure 11 — Deployment governance timeline (SOP-EPAPER-DEPLOY-001): the acceptance test is a hard gate at T-7 — no panel travels to site untested; burn-in alerting (3 missed heartbeats, over-temperature, UPS state) runs through the full week before handover.

11.5 Procurement-tracking data model (the §11.2 RFQ discipline as a schema)

The Appendix's procurement-tracking **ERD** gives the RFQ workflow a relational backbone — six entities chained `suppliers` → `rfqs` → `quotes` → `purchase_orders`, with quotes referencing panels and panels linked to their compatible `driver_boards` (the §2.2 pairing rule, enforced as a foreign key):

ENTITY	PURPOSE	DECISION-CRITICAL FIELDS
suppliers	Vendor registry	region, channel (direct/distributor), support_quality
panels	Panel catalog	technology, temp_min_c/temp_max_c (drives the Figure-3 tree), interface (drives the board pairing)
driver_boards	Matched TCON/adaptor per panel	interface, panel_id (FK — no orphan boards)
rfqs	One row per supplier enquiry	spec_version — quotes are only comparable against the same content spec
quotes	The §11.2 mandatory line items, as columns	unit_price_usd, lead_time_days, moq, warranty_months, doa_policy, incoterms, peak_refresh_power_w , push_api_open
purchase_orders	Commitment & landed cost	customs_duty_pct (§12: ~5% into Qatar), landed_cost_usd, acceptance_test_passed

The two bold columns are the report's signature requirements made queryable: a supplier who cannot fill `peak_refresh_power_w` blocks the §9 UPS sizing, and `push_api_open = false` flags a vendor-locked unit that breaks the §7 content interface. Open items carried in the TDD: confirm the actual Qatar customs duty rate with Finance, IP65 availability of GDEP312TT3 from the preferred supplier, and MTBF targets for the acceptance criteria.

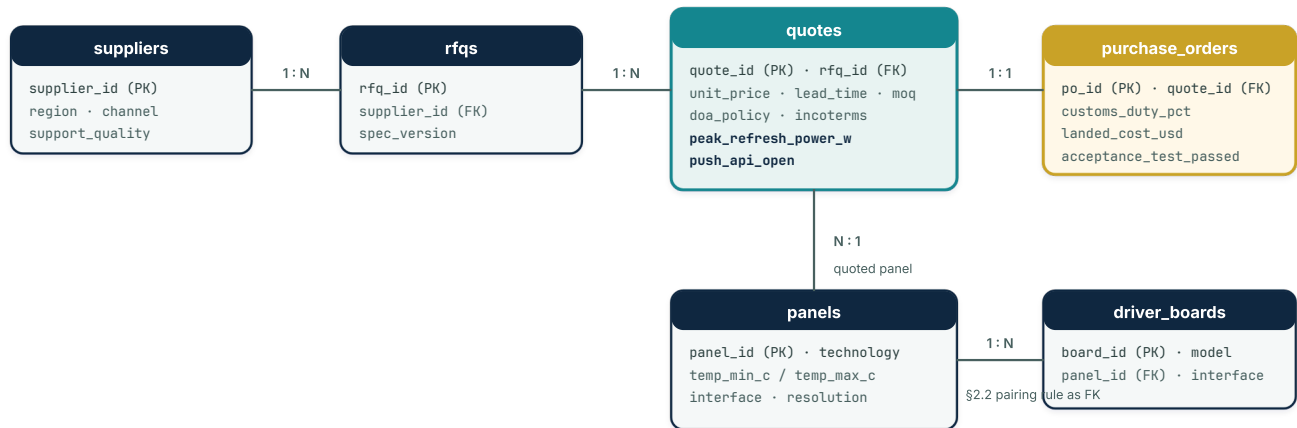


Figure 12 — Procurement-tracking ERD (from the project TDD): the §11.2 RFQ discipline as a schema; bold fields in quotes are the two report-signature requirements — missing `peak_refresh_power_w` blocks UPS sizing, and `push_api_open = false` flags a vendor-locked unit.

SECTION 12

Engineering Considerations for Qatar

12

- **Enclosure & shading (outdoor):** even -15/65°C panels need an IP65 enclosure + sun-shade + anti-glare; direct Doha sun can push the surface above 65°C.
- **Do not mix technologies:** Spectra 6 = indoor only; Kaleido 3 Outdoor / Marquee / ChLCD = outdoor.
- **Refresh speed:** slows in high heat — account for it in your content cadence.
- **ESD (dry climate):** Qatar's low indoor humidity (<30% RH) raises ESD risk during FPC/panel handling — ground straps and a written handling protocol are mandatory at the integration bench (§11.3).
- **Ghosting prevention:** rotate content (≥ 2 alternating images) on long-static deployments; verify ghosting in the delivery acceptance test (§11.3).
- **Power:** 0W static suits solar/battery installation with no permanent power cable.
- **Customs:** ~5% customs duty on imports (no VAT yet) — budget for it in the landed cost.
- **Sourcing channels:** Good Display via good-display.com / buyepaper.com / Alibaba; E Ink directly via shopkits.eink.com; Kaleido Outdoor and Marquee by contacting E Ink or its partners; ChLCD via IRIS Optronics; small panels and HATs via Waveshare/AliExpress/Amazon.

Security considerations (from the project TDD)

- **Physical:** intrusion sensors on the kiosk enclosure; ESD discipline at every FPC touch; IP65 sealing verified on outdoor installs (gasket check is an SOP step, not an assumption).
- **Network:** the kiosk WiFi AP lives on an isolated VLAN; the system is fully offline by design (TDD non-goal NG2), so there is no cloud attack surface; admin access requires physical presence + authentication.
- **Data integrity:** CRC-32 on every image transfer before any refresh; heartbeat monitoring bounds TCON-failure detection at 90 s; every content update is audit-logged with timestamp and checksum.
- **Supply chain:** supplier vetting recorded in the §11.5 tracking model; 100% delivery acceptance testing; ESP32 TCON firmware signed and verified at boot.

SECTION 13

Final Recommendations (Scenario Recipes)

13

SCENARIO	SUGGESTED RECIPE	CONTROLLER / CONNECTION
Indoor color signage/menu ~32"	Good Display 31.5" Spectra 6 + ESP32-S3 board	ESP32 (WiFi) or Android over WiFi
Cheap color prototype	Waveshare 13.3" E6 + ESP32-S3	ESP32 / RPi
Integration with your own MCU (small)	Bare panel ≤13.3" + DESPI	RPi (40-pin) / STM32 (SPI)
Outdoor color signage (Doha)	Kaleido 3 Outdoor 25.3" + Luvia + IP65 enclosure	E Ink board + shading
Outdoor, max color robustness/sun	ChLCD (IRIS Optronics)	B2B solution
Economical outdoor (color not needed)	GDEP312TT3 (−15/65°C), or IP65 V32-OSM / DMPH312EC1	IT8951 / built-in (USB/WiFi)
Your kiosk (RPi5 + Android + ESP32)	Panel + TCON as controller; RPi5/Android as content source	WiFi/USB to the TCON

SECTION 14

Reference: Benchmark Display & Market Context

14

Benchmark display — Samsung Color E-Paper (EM32DX): 32", QHD 2560×1440, based on E Ink Spectra 6, weighs 2.5 kg (with battery), 17.9 mm thick, 0.00W static consumption (per IEC62301: anything below 0.005W is reported as 0.00W), launched June 2025 at ~\$1,080. Every alternative in this report is also 0W static because it is genuine bistable e-paper.

Market context: E Ink Holdings is the world's largest ePaper display supplier (around 55–60% of the market), and most Chinese manufacturers are converters that depend on it (the notable exception is OED, with its own film). Spectra 6 won SID's Display of the Year award, and E Ink is building a new fab (H6) in Taoyuan at roughly \$107 million to scale up large-format panels.

This comprehensive report is based on official product pages and public sources as of June 2026. Prices are estimates and subject to change; confirm price, package contents (panel and board are usually separate), and temperature range directly with the supplier before purchasing.

Revision Notes (June 2026) — v10 Enhanced Edition

v10 delivers three of the Appendix's external diagram masters as native in-report figures, drawn in the report's own visual language directly from the companion documents: **Figure 10** (§7) — the full kiosk system architecture from TDD §3.1, showing the HMI / face-recognition / content-renderer layer over the local MQTT broker, the ESP32-S3 relay → door-lock chain, the display subsystem, and the UPS scoped to the critical path only; **Figure 11** (§11.4) — the SOP-EPAPER-DEPLOY-001 T-day governance timeline with the T-7 acceptance-test hard gate; and **Figure 12** (§11.5) — the procurement-tracking ERD with the two report-signature RFQ fields highlighted. Appendix B updated to mark the delivered masters. All v9 content and features are preserved.

v9 folds the project's two companion engineering documents into the report, turning architectural guidance into implementable specification: **§2.3** design goals & non-goals (G1-G6 / NG1-NG4) from the TDD; **§7.5** the content-push wire protocol made formal (HTTP POST /display/update with CRC headers, 200/400/503 semantics, MQTT display/status and display/heartbeat payloads, 30 s / 3-missed staleness rules); **§11.3** extended with the hardware qualification matrix (temperature cycling, 10,000-cycle life test, peak-power bench measurement, IP65 water-jet), concrete acceptance pass criteria (200 OK + CRC match, ±3°C temperature-sensor check, RMA-on-fail), and the 7-day burn-in protocol; **§11.4** the 16-week phased rollout with the week-8 readiness gate plus the SOP-EPAPER-DEPLOY-001 T-day governance timeline; **§11.5** the procurement-tracking data model whose quotes entity encodes the mandatory RFQ line items as queryable columns; and a **§12 security block** (physical / network / data-integrity / supply-chain). Glossary extended with MQTT, RMA, Burn-in, and MTBF. All v8 content and features are preserved.

v7–v8 are presentation and accessibility passes with all technical content preserved: interactive navigation with section tracking, a searchable 30+ term technical glossary with inline tooltips, dark mode, expandable sections, enhanced figures, full A4 print/PDF output (collapsed sections auto-expanded, repeated table headers, page-break control), responsive layout, and keyboard/touch accessibility throughout.

Appendix — Companion Diagram Pack (editable masters)

Every figure and process in this report is maintained as a live, editable master diagram in the project's internal workspace (Lucidchart and Eraser). All ten masters were visually verified on the live canvas — layout and arrow semantics — before publication; the editable files are available from the author (Eng. Abdulaziz N. AL-Haddad) on request.

A. Integration figures (= Figures 1–2, §7): the physical signal chain, and the kiosk content-push architecture.

B. Engineering & procurement set: ① *panel-selection decision tree* (= Figure 3, §11.1); ② *content-push sequence diagram* with the heartbeat loop and alarm branch (= Figure 4 — the delivered §11.2 P0 artifact); ③ *full kiosk system architecture* — **now delivered in-report as Figure 10** (§7); ④ *procurement-tracking ERD* — **now delivered in-report as Figure 12** (§11.5); ⑤ *three-lane procurement BPMN* (§11.2); ⑥ *deployment T-day timeline* — **delivered as Figure 11** (§11.4, from the SOP).

C. Application patterns set (= Figures 5–9, §7): ① USB/TF advertising poster; ② phone-updated restaurant menu (BLE / WiFi web server); ③ battery-free NFC labels; ④ city-wide 4G/CMS outdoor fleet with solar and telemetry; ⑤ self-generating live dashboard (the kiosk pattern).

D. Companion engineering documents (Eraser workspace): ① *Technical Design Document — E-Paper Display System for Doha Access Control Kiosks* (goals/non-goals, layered architecture, content-push protocol spec, procurement data model, testing strategy, 16-week rollout) — integrated into this report as §2.3, §7.5, §11.3–11.5 and the §12 security block; ② *SOP-EPAPER-DEPLOY-001 — E-Paper Display Kiosk Deployment* (roles, T-day timeline, delivery acceptance procedure, installation/ESD discipline, integration tests, burn-in, handover forms).

APPENDIX

Technical Glossary

A

Alphabetical reference of the 35 technical terms used throughout this report.

ACK — Acknowledgment signal - confirms successful receipt of data in the content-push handshake protocol.

Bistable — The property of e-paper where each pixel maintains its state without power - the image stays visible at 0W.

BOM — Bill of Materials - a comprehensive list of components, parts, and prices needed for a project.

Burn-in — A supervised initial operating period (7 days in the deployment SOP) with automated monitoring, used to surface early failures before handover to operations.

CMS — Content Management System - software for scheduling and pushing content to digital signage fleets remotely.

CRC — Cyclic Redundancy Check - an error-detecting code used to verify data integrity during image transfers.

DDP — Delivered Duty Paid - an Incoterm where the seller bears all costs and risks until the goods are delivered to the buyer.

DESPI — Good Display's SPI adapter board series for small e-paper panels (C02, C73, C1085, etc.).

DOA — Dead on Arrival - a product that is non-functional when first received. Critical for return policies on expensive panels.

EPD — Electrophoretic Display - the technical name for e-paper technology using charged pigment particles in fluid.

ESD — Electrostatic Discharge - a sudden flow of electricity that can damage sensitive electronic components like FPC connectors.

ESL — Electronic Shelf Label - small e-paper tags used in retail for dynamic pricing, often battery-free with NFC.

FPC — Flexible Printed Circuit - a thin, flexible ribbon cable that connects the e-paper panel to the driver board. Fragile and must be handled with care.

Framebuffer — A memory buffer storing the image to be displayed - large color panels require significant RAM (e.g., ESP32 over Arduino).

Ghosting — A faint residual image from previous content that remains visible after refresh - caused by incomplete particle movement.

Heartbeat — A periodic signal from the TCON to the host confirming the system is alive - used by the staleness watchdog.

Incoterms — International Commercial Terms - standardized trade terms defining buyer/seller responsibilities in global shipping.

IP65 — Ingress Protection rating: dust-tight and protected against water jets - essential for outdoor enclosures in Qatar.

IT8951 — A generic driver IC for grayscale Carta e-paper panels, supporting USB, SPI, and I80 interfaces.

Level Shifter — A circuit that converts logic levels (e.g., 5V Arduino to 3.3V panel) to prevent damage to the display.

LVDS — Low-Voltage Differential Signaling - a high-speed interface used by the 25.3" Kaleido 3 Outdoor panel.

MOQ — Minimum Order Quantity - the smallest number of units a supplier will sell in a single order.

MQTT — Message Queuing Telemetry Transport - a lightweight publish/subscribe messaging protocol. The kiosk runs a local broker on the RPi5 for heartbeats and status events, fully offline.

MTBF — Mean Time Between Failures - a reliability metric; target values feed the acceptance test criteria (open question in the TDD).

NDEF — NFC Data Exchange Format - a standardized format for storing data on NFC tags (e.g., vCards, URLs).

PMIC — Power Management Integrated Circuit - regulates the multiple voltage rails required by e-paper panels.

QSPI — Quad SPI - a faster variant of SPI using four data lines, required by large panels like the 31.5" Spectra 6.

RF — Radio Frequency - electromagnetic waves used by NFC to power and communicate with battery-free e-paper tags.

RH — Relative Humidity - Qatar's low indoor RH (<30%) increases ESD risk during panel handling.

RMA — Return Merchandise Authorization - the supplier's formal authorization to return a defective unit. Agree RMA/DOA logistics to Qatar in the RFQ before ordering large panels.

SPI — Serial Peripheral Interface - a synchronous serial communication protocol used by small e-paper panels (up to 13.3").

Staleness Watchdog — A monitoring mechanism that detects when the TCON stops updating, preventing outdated content from being displayed indefinitely.

TCON — Timing Controller - the chip on the driver board that generates the precise waveforms needed to update the e-paper display.

UPS — Uninterruptible Power Supply - a battery backup that keeps critical systems running during power outages.

Waveform — The specific voltage sequence applied to e-paper particles to move them between color states - unique per panel type.